



SOROTI UNIVERSITY
SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

UgaWildlifeVision: An Annotated Dataset and a Wildlife Detection and Recognition System for Animal Species in Uganda

GEC4101 Research Project Proposal Report

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By

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Statement of Declaration

I, BABIRYE AISHA, 2301600101, declare that this work is my own original effort and has not been submitted to any other institution for the award of a degree or any other academic credit. All sources of information used in this work have been duly acknowledged through complete references.

Signature: _____

Date: _____

Approval

This GEC4101 Research Project Proposal Report has been submitted for examination with the approval of the undersigned supervisor(s).

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Date: _____

Abstract

Uganda's tourism sector welcomed more than 1.37 million international visitors and earned about USD 1.28 billion in 2024, and wildlife viewing in the country's national parks is one of its main attractions. Most visitors, however, cannot reliably tell the species they photograph apart, they depend on a guide who may not always be available, and they receive no information in a local language. Existing wildlife AI tools target researchers using camera-trap images, are trained on data from other regions, and are not available in Luganda, and no open, bounding-box-annotated dataset of Ugandan wildlife in tourist-like conditions is known.

This proposal presents *UgaWildlifeVision*, a two-part project. The first part is an annotated image dataset of seven priority Ugandan wildlife classes (chimpanzee, gorilla, antelope, hippopotamus, elephant, bird and hog), built by extending an existing safari wildlife dataset with openly licensed images of Ugandan wildlife collected from online sources, with species-labelled bounding boxes for at least about 5,000 images. The second part is a mobile application that lets a tourist take or upload a photograph and receive, for each animal detected, a bounding box, the species name and a confidence score. The results are shown through an augmented-reality (AR) overlay and are accompanied by a written Luganda description of the result.

A design-science approach will be followed. Lightweight object detectors from the YOLO family and mobile-oriented baselines will be fine-tuned on the dataset, compared, and exported to TensorFlow Lite so that detection runs on the phone without an internet connection. A curated, linguist-validated Luganda knowledge base will generate the Luganda text. The system will be evaluated technically (mean average precision, latency, calibration, robustness) and with users (System Usability Scale, task success, Luganda comprehension).

The expected outcomes are an open, documented dataset, a benchmarked on-device detection model, and a working bilingual AR application. These should improve the visitor experience, support conservation education and extend language technology to Luganda.

Keywords: wildlife detection; object detection; augmented reality; Luganda; annotated dataset; mobile application; Uganda

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List of Abbreviations

AI	Artificial Intelligence
AP	Average Precision
API	Application Programming Interface
AR	Augmented Reality
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CVAT	Computer Vision Annotation Tool
ECE	Expected Calibration Error
GPU	Graphics Processing Unit
IoU	Intersection over Union
mAP	mean Average Precision
ML	Machine Learning
MT	Machine Translation
NLLB	No Language Left Behind
NMS	Non-Maximum Suppression
RQ	Research Question
SALT	Sunbird African Language Technology
SDLC	Software Development Life Cycle
SUS	System Usability Scale
TAM	Technology Acceptance Model
TFLite	TensorFlow Lite
UGX	Uganda Shillings
UWA	Uganda Wildlife Authority
UWEC	Uganda Wildlife Education Centre
YOLO	You Only Look Once

Chapter 1 Introduction

This chapter presents the context, problem, objectives, research questions, justification, significance and scope of *UgaWildlifeVision*.

1.1 Background

Uganda is one of Africa's most biodiverse countries, and wildlife viewing in parks managed by the Uganda Wildlife Authority (UWA), such as Murchison Falls, Queen Elizabeth, Kibale and Bwindi, is a major tourist attraction. In 2024 the country received 1,371,895 international visitors, tourism receipts rose by 25.9% to about USD 1.28 billion, tourism contributed 3.2% of gross domestic product, and the sector supported more than 803,000 jobs [1]. At the same time wildlife is under pressure: lion numbers fell from 373 to about 292 [2], and some species recover while others remain under threat [3]. Visitors who understand the animals they see are more likely to support conservation, so wildlife tourism has an educational role as well as an economic one.

In practice, many visitors cannot name the animals they photograph, guides cannot accompany every visitor at all times, and information is mostly available only in English. Local visitors, students and communities, many of whom prefer Luganda, are therefore often left out.

Deep learning has transformed wildlife image analysis. Networks trained on millions of Snapshot Serengeti camera-trap images identified species with more than 93% accuracy and matched human volunteers [4]. Single-stage detectors from the YOLO family locate and classify objects in one pass and have small variants light enough for a phone [5, 6], and smartphone augmented reality (AR) can place information over the live camera view [7, 8]. What is missing is a dataset and a system that apply these technologies to Ugandan wildlife, Ugandan conditions and the Luganda language.

1.2 Problem Statement

Tourists in Uganda's parks frequently cannot identify the animals they photograph, and the available tools do not fit the Ugandan context, for four reasons. First, no openly available,

bounding-box-annotated image dataset of Ugandan wildlife photographed from a visitor's viewpoint is known to the author; African datasets such as Snapshot Serengeti and Snapshot Safari consist of fixed camera-trap images, mostly from Tanzania and southern Africa [9, 10], and models trained on such data generalise poorly to new locations and viewpoints [11]. Second, existing tools are aimed at researchers or are general consumer applications [12] that, to the best of the author's knowledge, do not combine a bounding box, a confidence score and AR information for Ugandan wildlife. Third, Luganda is a low-resource language with few digital resources [13, 14], so wildlife information in Luganda is practically unavailable. Fourth, park areas have patchy connectivity and only about one in three Ugandan mobile subscribers uses a smartphone [15], so cloud-dependent solutions are unreliable. As a result, visitors miss learning opportunities, guides are under-supported, and Luganda speakers are excluded from digital wildlife education.

1.3 Objectives of the Study

1.3.1 General Objective

To develop an annotated image dataset of Ugandan wildlife and a mobile detection and recognition system that, from a tourist's photograph, shows each animal's bounding box, species name and confidence score through augmented reality, together with a Luganda text description of the result.

1.3.2 Specific Objectives

- (i) To extend the existing dataset with new images collected from online sources to cover seven classes (chimpanzee, gorilla, antelope, hippopotamus, elephant, bird and hog), with at least about 700 annotated images per class, and to document it with a datasheet.
- (ii) To train, compare and select a lightweight detector, targeting mAP@0.5 of at least 85% on a held-out test set and under 300 ms per image on a mid-range Android phone.
- (iii) To develop a Luganda results module that gives a validated Luganda description of each result.
- (iv) To develop a mobile application that accepts a captured or uploaded photograph and presents the bounding box, species name, confidence score and Luganda text in an AR

overlay.

- (v) To evaluate the dataset, model and application technically and with users, including a System Usability Scale (SUS) score of at least 68.

1.4 Research Questions

RQ1: What image volume and annotation protocol give a representative, high-quality dataset of Ugandan wildlife in tourist-style conditions? (Objective i)

RQ2: Which lightweight detector gives the best balance of accuracy, calibration and speed on mid-range phones, and how well does it generalise to unseen locations? (Objective ii)

RQ3: How can detection results be rendered in correct, understandable Luganda? (Objective iii)

RQ4: How can boxes, names and scores be presented in a stable, readable AR view in field conditions? (Objective iv)

RQ5: How do tourists and guides rate the usability, usefulness and trustworthiness of the system? (Objective v)

1.5 Justification of the Study

The study is necessary because the data and tools that would let AI help visitors in Ugandan parks do not exist in a usable form, and models from other regions or camera-trap viewpoints do not transfer reliably [11]. The timing is favourable: tourism is growing [1], wildlife needs public support [3], efficient mobile detectors and AR frameworks are now cheap and available [16, 8], and recent resources have begun to cover Luganda [13, 14]. Placing Luganda in a working application also contributes directly to closing a digital-inclusion gap.

1.6 Significance of the Study

Tourists get immediate, offline identification with a transparent confidence score. **Local visitors, schools and communities** get wildlife information in Luganda. **extbfGuides** get a tool to verify and enrich tours. **UWA and researchers** gain a documented dataset and a

new East African benchmark that combines computer vision with a low-resource African language.

1.7 Scope of the Study

1.7.1 Content/Technical Scope

The project covers: (a) extending the existing dataset with new annotated images of seven classes (chimpanzee, gorilla, antelope, hippopotamus, elephant, bird and hog), where antelope, bird and hog are *group classes* containing several species (Appendix D); (b) training, benchmarking and exporting a lightweight detector to TensorFlow Lite; (c) an Android application with image capture and upload, bounding boxes, species name, confidence score, AR overlay and Luganda text; (d) a validated Luganda knowledge base; and (e) technical and user evaluation. It does not cover species-level identification within the group classes, other animal groups or plants, video tracking and counting, behaviour recognition, poaching detection, iOS, or Luganda speech output.

1.7.2 Geographical Scope

The study concerns wildlife in Uganda. No field trip is planned: all new data will be collected online, from repositories and platforms that hold images of Ugandan wildlife, selected by species and, where available, by observation location in Uganda. Where too few Ugandan images exist for a class (for example gorilla), the existing dataset and openly licensed East African images will be used. Participants will be recruited in Uganda.

1.7.3 Time Scope

The project runs for about eight months, from November 2026 to June 2027 (Appendix A).

1.8 Organization of this Report

Chapter 2 reviews related literature and presents the research gap and frameworks. Chapter 3 describes the methodology, architecture, requirements, evaluation, ethics and risks. Chapter 4 concludes. The appendices give the work plan, the budget in Uganda Shillings, the data collection instruments and technical information.

Chapter 2 Literature Review

This chapter reviews the key concepts, existing systems, the research gap and the frameworks.

2.1 Review of Relevant Concepts

Object detection. Classification gives one label per image, whereas detection finds each object, draws a bounding box and gives a class label and confidence score. A photograph may contain several animals, and the box shows which one is identified. Single-stage detectors such as YOLO predict boxes and classes in one pass and made real-time detection practical [5]; YOLO11 offers nano to extra-large variants, the small ones suited to phones [17, 6], and MobileNet-style networks use depthwise-separable convolutions to cut computation [16]. Quality is measured by Intersection over Union (IoU), precision, recall and mean average precision (mAP), following the PASCAL VOC and COCO conventions [18, 19].

Transfer learning. Fine-tuning a pre-trained detector needs far less data than training from scratch, because early layers learn generic features [20].

Domain shift and dataset design. Most wildlife deep learning uses camera-trap images [4, 21], but models that perform well where they were trained often perform much worse at new locations [11]. Tourist photographs differ from camera-trap frames in viewpoint, distance and framing, so the dataset must contain such images and testing must hold out whole locations. Good practice also requires annotation guidelines, annotator agreement checks such as Cohen's kappa [22], and a datasheet [23].

On-device inference and calibration. Running the model on the phone removes the need for a network, protects privacy and reduces delay; TensorFlow Lite (LiteRT) supports this, with quantisation shrinking the model at small accuracy cost [24]. A confidence score is useful to tourists only if it is calibrated, that is, detections scored 0.8 are correct about 80% of the time, so calibration will be measured and low-confidence detections reported as uncertain.

Augmented reality. AR combines real and virtual content, registered in three dimensions and interactive in real time [7]. ARCore keeps virtual labels attached to the scene as the

camera moves [8].

Luganda language technology. Luganda is a widely spoken Bantu language that is low-resource in NLP. The SALT corpus and models from Makerere AI Lab and Sunbird AI reach BLEU of about 26 into English and 20 out of English [13, 25], and Meta’s NLLB-200 includes Luganda [14]. For a short, fixed vocabulary, a curated knowledge base checked by native speakers is more dependable than free translation.

2.2 Review of Existing Systems, Technologies and Approaches

Norouzzadeh *et al.* trained CNNs on 3.2 million Snapshot Serengeti images and reached over 93% species accuracy, automating 99.3% of the labelling at human-level accuracy [4]. MLWIC reached 98% accuracy on US data but only 82% on out-of-sample Canadian data [26]. MegaDetector finds animals, people and vehicles in camera-trap images but does not name the species [27], and Wildlife Insights offers cloud-based species identification for camera-trap managers [28]. The Caltech Camera Traps benchmark showed that generalisation to unseen locations is poor [11]. The Ultralytics African Wildlife set has only four classes and about 1,500 images [29]. Consumer applications such as Seek identify organisms from a global taxonomy [12], but to the best of the author’s knowledge none shows a calibrated confidence with a bounding box for Ugandan wildlife, or Luganda output. Table 2.1 summarises the comparison.

Table 2.1: Comparison of reviewed approaches with the proposed system.

System	User	Output	On phone	Uganda	Luganda
Serengeti CNNs [4]	Researchers	Species, count	No	No	No
MLWIC [26]	Researchers	Species	No	No	No
MegaDetector [27]	Researchers	Animal box	No	No	No
Wildlife Insights [28]	Researchers	Species (cloud)	No	Partial	No
Seek [12]	Public	Species (global)	Partial	No	No
UgaWildlifeVision	Tourists	Box, species, score, AR	Yes	Yes	Yes

2.3 Research Gap

The review shows that deep learning identifies African wildlife accurately and that mobile detectors and AR are mature, but it reveals four gaps: (1) **data**, since no open, annotated, multi-class dataset of Ugandan wildlife with tourist-style images is known, and models from other regions generalise poorly [11]; (2) **system**, since existing tools are researcher- and cloud-oriented and none gives an offline, visitor-facing AR identification for Ugandan wildlife; (3) **language**, since no wildlife or computer-vision application is known to present results in Luganda; and (4) **evaluation**, since few studies assess calibration, on-device speed and usability together. This project addresses all four.

2.4 Theoretical Framework

The **Technology Acceptance Model (TAM)** holds that intention to use a technology is determined mainly by perceived usefulness and ease of use [30], with enjoyment and information quality added for mobile AR in tourism [31]. It guides what the user study measures: usefulness, ease of use, trust in the confidence score and understanding of the Luganda text. **Transfer learning** [20] justifies fine-tuning a pre-trained detector and explains why the quality and representativeness of the target data decide performance. The development process follows Design Science Research (Section 3.1) [32, 33].

2.5 Conceptual Framework

Figure 2.1 proposes that *dataset quality* (coverage, annotation quality, balance) determines *detection performance* (accuracy, calibration, speed); performance and the *quality of result presentation* (AR clarity, Luganda text) shape the tourist's *perceived usefulness, ease of use and trust*, which influence *intention to use and conservation learning*. *Capture conditions* (distance, light, occlusion) and *device capability* moderate performance and experience. The framework links to RQ1–RQ5.

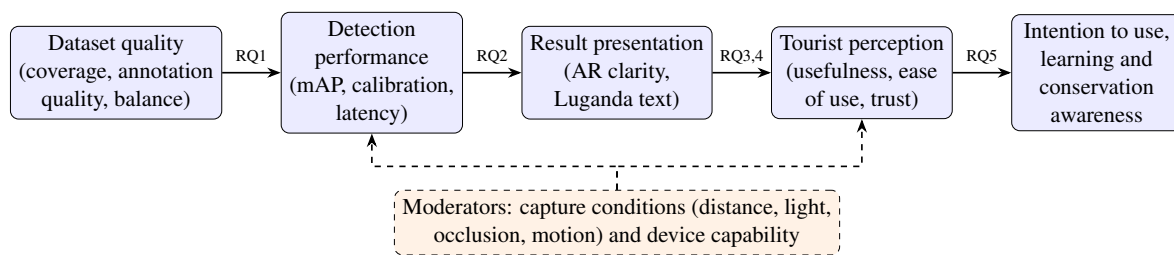


Figure 2.1: Conceptual framework of the study.

Chapter 3 Proposed Methodology

This chapter describes the research design, architecture, data collection, tools, requirements, evaluation, ethics and risks.

3.1 Research Design

The project follows **Design Science Research (DSR)**, which suits work that creates and evaluates artefacts (here a dataset, a detection model and a mobile application) to solve a practical problem [32]. The six DSR activities of Peffers *et al.* [33] map to the project as follows: problem identification (Chapter 1); objectives of a solution (Section 1.3); design and development (dataset, model, Luganda module, application); demonstration (running the application on tourist-style test photographs on real phones); evaluation (technical and user evaluation); and communication (final report and dataset release).

3.2 System Architecture

Figure 3.1 shows the proposed architecture. It has an *offline development pipeline*, which produces the dataset and the trained model, and an *on-device mobile application*, which uses the model to serve the tourist without a server or internet connection.

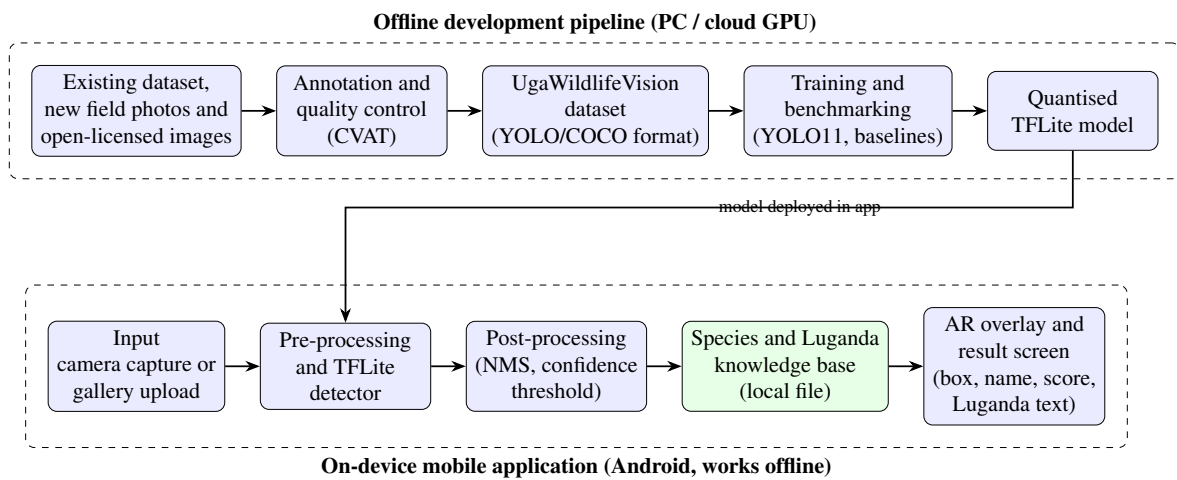


Figure 3.1: Proposed system architecture of UgaWildlifeVision.

In the pipeline, images are annotated with bounding boxes in CVAT [34], COCO-pre-

trained detectors are fine-tuned and compared, and the selected model is quantised and exported to TensorFlow Lite [24]. In the application, the *input module* takes a captured or uploaded photograph; the *detector* resizes it and outputs boxes, classes and scores; *post-processing* applies non-maximum suppression (NMS) and a confidence threshold, so low-confidence detections are shown as “uncertain” with alternatives; and the *knowledge base*, a local database of fact sheets and validated Luganda sentence templates, supplies the Luganda text for the detected class and confidence level. The *AR layer* draws the box, name, score and Luganda text over the camera view or photograph; with ARCore the information cards stay anchored to the animal as the phone moves [8], and phones without ARCore support get a two-dimensional overlay.

3.3 System Development Methodology

An **iterative, prototype-based Agile approach** with two-week sprints will be used, because requirements will be refined as the data audit and user feedback arrive and the model and application can be improved independently. The stages are planning and requirements, data collection and annotation, model development, application development, integration and testing, user evaluation and refinement, and release and documentation, matching the work plan (Appendix A).

3.4 Requirements/Specifications Elicitation

Requirements will come from a review of existing tools and UWA information; structured interviews with 6–8 tour guides or rangers on commonly asked species and current identification practice; a short questionnaire for about 30 tourists and Ugandan visitors on devices, connectivity, language and expectations; a review of sample tourist photographs on online platforms to see how animals are framed; and consultation with a Luganda language specialist and the supervisors. The findings will be converted into the requirements in Section 3.8 and refined in the first sprint (draft instruments in Appendix C).

3.5 Data Collection Methods

Image data. The existing safari wildlife dataset is the starting point. It will be audited (images and boxes per class, formats, sources, licences, duplicates) and converted to a com-

mon format, and the gaps found will set how many new images are needed. The target is seven classes (chimpanzee, gorilla, antelope, hippopotamus, elephant, bird and hog) with at least about 700 images per class, roughly 5,000 images and 8,000 boxes in total. No field trip is planned, so all new images will be collected from online sources: (a) **citizen-science and community photo repositories**, namely iNaturalist research-grade observations [35], Wikimedia Commons and Flickr, filtered by class, by observation location in Uganda where available, and by permissive licences (CC0, CC BY, CC BY-SA); these images are mostly taken by visitors with handheld cameras and phones, so they resemble the tourist-style photographs the application will receive; (b) public camera-trap collections such as Snapshot Serengeti and Snapshot Safari [9, 10], used to add East African variety, mainly for classes with few Ugandan images; and (c) images from UWA or tour operators only where their licence or written permission allows. Images will be downloaded through official APIs and download tools within each site's terms of use, and the source, URL, photographer, licence and observation location of every image will be recorded in a metadata file.

Annotation and splitting. Images are annotated in CVAT with tight boxes following a written guideline, and each image carries metadata (location, date, time of day, distance class, device, source, licence). Two annotators will label a 10% sample independently and agreement will be measured by IoU and Cohen's kappa [22]. The test set will hold out entire sources, photographers and observation locations, so that near-duplicate images are not shared between training and test sets and generalisation can be measured [11], with roughly 70% training, 15% validation and 15% test.

3.6 Tools, Materials and Technologies

Luganda output will not rely on free translation: a closed set of templates and fact sheets will be drafted in English, translated with model help and approved by a Luganda language specialist, so the application only fills a template, which is fast, predictable and offline.

3.7 Data Analysis Methods

Dataset statistics (images and boxes per class, distance, light, location) will show coverage and balance. Annotation quality will be reported as mean IoU and Cohen's kappa. Detection performance will be reported as precision, recall, per-class AP, mAP@0.5 and

Table 3.1: Proposed tools and technologies.

Tool	Purpose
Android smartphones	Application testing (the student’s ARCore-capable phone and a borrowed low-end phone).
Python download scripts, iNaturalist/Wikimedia/Flickr APIs	Collection of online images and their metadata.
CVAT [34]	Bounding-box annotation with YOLO and COCO export.
Python, PyTorch, Ultralytics YOLO11 [17]	Training and evaluation with transfer learning.
YOLOv8n, SSD-MobileNet baselines [16]	Comparison of accuracy, size and speed.
Google Colab / Kaggle GPU	Affordable cloud training.
TensorFlow Lite [24]	Quantisation and on-device inference.
Flutter / Kotlin and ARCore [8]	Application and AR overlay.
SQLite / JSON	Local species and Luganda knowledge base.
NLLB and SALT models [14, 25]	Drafting aid only; all final Luganda text is validated by a native speaker.
Git, Google Forms, Figma	Version control, questionnaires, interface mock-ups.

mAP@0.5:0.95 [18, 19], with confusion matrices and bootstrap confidence intervals. Calibration will be checked with reliability diagrams and expected calibration error (ECE), with temperature scaling if needed. Robustness will be analysed by distance, occlusion, light and unseen location, and efficiency by model size and latency on the test phones. Luganda quality will be summarised as native-speaker Likert ratings of adequacy and fluency, and user data as SUS scores [36], descriptive statistics of TAM items [30], task success and time, and thematic analysis of comments.

3.8 Preliminary Project Requirements and Specifications

3.8.1 Functional Requirements/Specifications

3.8.2 Non-Functional Requirements/Specifications

3.9 Proposed Testing and Evaluation Strategy

Dataset: completeness of labels, annotation agreement, class balance and absence of duplicates across splits, published with a datasheet [23]. **Model:** candidate detectors are compared on the held-out test set by mAP, per-class AP, precision, recall, confusion ma-

Table 3.2: Preliminary functional requirements.

ID	The system shall...	Objective
FR1	take a photograph with the in-app camera or accept one uploaded from the gallery.	(iv)
FR2	detect each supported animal and draw a bounding box around it.	(ii), (iv)
FR3	display the class (species) name and a confidence score for each detection.	(ii), (iv)
FR4	flag detections below a confidence threshold as “uncertain” and show the likely alternatives.	(ii)
FR5	present results in an AR overlay, with a two-dimensional fallback where AR is unsupported.	(iv)
FR6	display a written Luganda description of each result, including the confidence level.	(iii)
FR7	provide a short fact sheet per class in English and Luganda, and switch the interface language.	(iii)
FR8	perform detection without an internet connection and keep a local, deletable history.	(ii), (iv)

trix, calibration, size and latency before and after quantisation, with robustness subsets (distance, light, occlusion, unseen location) and error analysis of failures such as look-alike antelopes, small distant birds and animals hidden in vegetation. **Application:** functional tests for each requirement (including offline operation and the non-AR fallback); AR tests of overlay alignment and stability; a device test on held-out tourist-style images and live camera input, recording correct identifications, response time and problems; and a usability test with about 20 participants (tourists, Ugandan visitors, guides, students) who complete defined tasks and answer the SUS and TAM questionnaire, while Luganda speakers rate the Luganda text.

3.10 Ethical and Social Considerations

3.10.1 Ethical Considerations

Interviews, questionnaires and usability tests involve human participants, so participation will be voluntary with written informed consent (Appendix C), the right to withdraw, anonymised and securely stored responses, and ethics clearance sought before data collection, in line with the Data Protection and Privacy Act, 2019 [37]. People incidentally photographed will be avoided, blurred or removed. Because no field work is planned, no animal is disturbed. Images will be downloaded only from sources whose terms of use allow it, only images with permissive licences will be used, and the photographer, source and licence of

Table 3.3: Preliminary non-functional requirements and targets.

Attribute	Target specification
Accuracy	mAP@0.5 of at least 85% on the held-out test set.
Performance	At most 300 ms per photograph on a mid-range Android phone; at least 10 frames per second in live preview.
Calibration	Expected calibration error of at most 0.10.
Usability	SUS score of at least 68; task success rate of at least 90% for first-time users.
Language quality	Mean native-speaker rating of at least 4 out of 5 for Luganda text.
Portability and size	Android 9 or above; application and model of about 100 MB or less.
Reliability	No crash in a 30-minute session; clear message when no animal is found.
Privacy	Images processed locally; no personal data sent without consent, in line with the Data Protection and Privacy Act, 2019 [37].

every image will be recorded and credited. To limit dual-use risk, GPS data will be removed and only park-level location kept in the released dataset, and the application will not publish animal locations. The application will state its limitations and show uncertainty so that it is not treated as an authority.

3.10.2 Social Considerations

Offering Luganda widens access for Ugandan visitors, schools and communities, but other languages are not covered and many people lack smartphones [15]; the app will therefore be offline, light and extensible to further languages, and its Luganda will be checked by native speakers. It is meant to support, not replace, tour guides, who will be consulted. The interface will remind users to keep a safe, lawful distance from animals. Release terms and credit to the original photographers, platforms and, where possible, UWA will be agreed in advance so that the benefits of the Ugandan data stay with Ugandan stakeholders.

3.11 Risk Analysis

3.12 Chapter Summary

The project follows Design Science Research to build a dataset, a lightweight detector and a bilingual AR application. An offline pipeline produces the model, and an on-device application serves the tourist without internet access. Requirements come from guides, rangers and visitors, data from the existing dataset plus new images collected online, and evaluation

Table 3.4: Risk register.

Risk	Likelihood Impact		Mitigation
Too few openly licensed images from Uganda for some classes	Medium	High	Search several sources (iNaturalist, Wikimedia Commons, Flickr, camera-trap collections); use the existing dataset; accept East African images for group classes and state the limits.
Mislabeled, duplicate or poor-quality online images	High	Medium	Manual review of every image, duplicate removal and label checks by a second annotator.
Accuracy below target; confusion inside antelope and bird group classes	Medium	High	Collect more data for weak classes, augment, try larger variants, report per-class results.
Poor generalisation to unseen locations	Medium	High	Hold out locations; collect at varied sites, times and light; add hard negatives.
Model too slow for low-end phones	Medium	Medium	Use nano/small variants and quantisation; benchmark early.
AR unsupported or unstable on some devices	Medium	Medium	Two-dimensional fallback; test on two devices.
Inaccurate Luganda text	Medium	High	Closed templates reviewed by a native speaker; release no unreviewed text.
Budget overrun or schedule slippage	Medium	Medium	Contingency fund; free tiers first; weekly supervisor check-ins.

covers accuracy, calibration, speed and usability, with ethics and risks addressed.

Chapter 4 Conclusion

This proposal has presented a plan to address the problem that tourists and other visitors to Uganda's parks cannot easily identify the animals they photograph, because existing artificial-intelligence tools are designed for researchers, trained on data from other regions and viewpoints, dependent on connectivity, and unavailable in Luganda. The study is motivated by the importance of wildlife tourism to the Ugandan economy and by the role visitor understanding can play in conservation. The literature review established that deep learning is accurate for African wildlife, but that no open annotated dataset of Ugandan wildlife in tourist-style imagery, and no offline, localised, AR-based identification application, is known.

The proposed system, UgaWildlifeVision (Section 3.2), combines an annotated dataset of seven Ugandan wildlife classes (chimpanzee, gorilla, antelope, hippopotamus, elephant, bird and hog) with an on-device mobile application that shows each detected animal's bounding box, species name and confidence score in augmented reality, with a validated Luganda description. The methodology in Chapter 3 gives a structured and feasible path to the objectives in Chapter 1, and the project is achievable within the time and budget in Appendices A and B.

Upon completion, the project is expected to contribute an open, documented dataset, a benchmarked lightweight detector and a working bilingual AR application that improves the visitor experience, supports guides and conservation education, and extends language technology to Luganda. The student is committed to the ethical and social responsibilities identified here and to working under the guidance of the supervisors to deliver a high-quality final-year project.

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Appendix A Project Workplan

The project runs for eight months, from November 2026 to June 2027; dates are indicative and will be aligned with the departmental calendar.

Table A.1: Work plan: activities, timeline and expected outputs.

No.	Activity	Period	Months	Output
1	Proposal defence and ethics clearance	Nov 2026	1	Approved proposal, ethics clearance
2	Requirements elicitation	Nov–Dec 2026	2	Requirements baseline
3	Dataset audit and online data collection	Nov 2026–Feb 2027	4	Audited dataset, new on-line images
4	Annotation and quality control	Dec 2026–Mar 2027	4	Frozen annotated dataset
5	Luganda knowledge base and validation	Jan–Mar 2027	3	Validated Luganda text
6	Model training, benchmarking, TFLite export	Feb–Apr 2027	3	Selected quantised model
7	Mobile and AR application development	Feb–May 2027	4	Working application
8	Integration, testing and user evaluation	Apr–Jun 2027	3	Evaluation results
9	Dataset release and documentation	May–Jun 2027	2	Datasheet, public release
10	Final report and presentation	May–Jun 2027	2	Final report

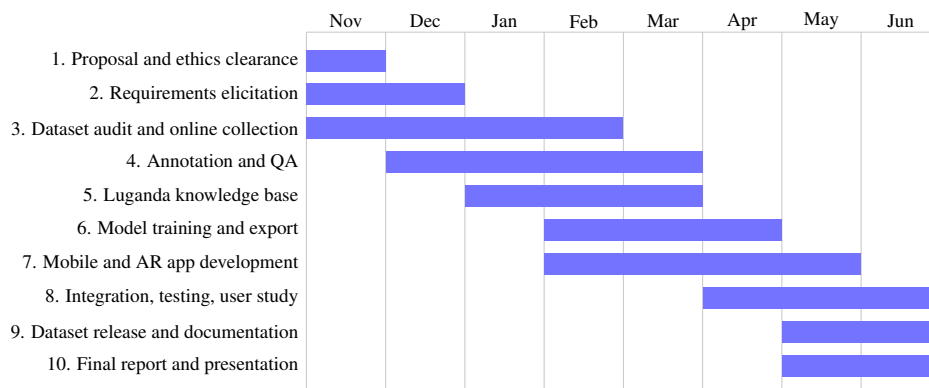


Figure A.1: Gantt chart of the work plan (Month 1 = November 2026).

Key milestones: M1 (end Dec 2026) requirements baseline and ethics clearance in place, existing dataset audited; M2 (end Feb 2027) dataset version 1 (at least 3,500 images annotated) and baseline detector; M3 (end Mar 2027) dataset frozen (at least 5,000 images) and Luganda knowledge base validated; M4 (end Apr 2027) model selected and exported to TFLite; M5 (end May 2027) working AR application prototype; M6 (end Jun 2027) user evaluation complete, dataset released and final report submitted.

Appendix B Budget and Budget Justification

All amounts are estimates in Uganda Shillings (UGX) to be confirmed against current prices. The total is kept to UGX 850,000 by using the student’s own laptop and smartphone, free software and free cloud tiers; no field trip is needed because the project starts from an existing safari wildlife dataset and collects all new data online.

Table B.1: Estimated budget in Uganda Shillings (UGX).

No.	Item	Qty	Unit cost	Total
1	Cloud GPU subscription (months)	3	45,000	135,000
2	Mobile data and internet for downloads and other uses (months)	5	30,000	150,000
3	Cloud storage for the dataset and back-ups (months)	3	10,000	30,000
4	Luganda language specialist honorarium	1	100,000	100,000
5	Annotation assistants (persons)	2	75,000	150,000
6	User-testing incentives (participants)	20	5,000	100,000
7	Printing and binding (copies)	4	20,000	80,000
Subtotal				745,000
Contingency (about 14%)				105,000
Grand total				850,000

Justification. Because the project starts from an existing dataset and collects new data online, the main costs are computing and connectivity. Cloud GPU (item 1) is much cheaper than buying a GPU, and the free Colab and Kaggle tiers will be used first. Mobile data and internet (2) are needed to download the existing dataset and the new images with their metadata, to train on cloud GPUs, to hold meetings with the supervisors and the language specialist, and to run the surveys. Cloud storage (3) holds the dataset and a backup copy. The language specialist (4) is the most reliable quality control for Objective (iii). Because the online images must be reviewed one by one for quality, labels and duplicates, two annotation assistants (5) share the annotation of about 8,000 boxes and provide the second annotations needed for agreement checks (Objective (i)). User-testing incentives (6) support Objective (v), and printing (7) covers the report. The student’s own laptop and ARCore-capable smartphone are used for development and testing, a borrowed low-end phone is used for performance tests, the application is installed directly (side-loaded) without a store account, and open-source software (Python, PyTorch, Ultralytics, CVAT, TensorFlow Lite, Flutter) is free, so no hardware or licence costs are included.

Appendix C Data Collection Instruments

Drafts, to be refined with the supervisors and submitted for ethics clearance.

Informed consent statement. *You are invited to take part in a study by a final-year student of Soroti University (GEC4101) on a mobile application that identifies animals from photographs and shows the result in augmented reality and in Luganda. Taking part involves answering questions and/or trying the application for about 20–30 minutes. Participation is voluntary, you may skip any question or withdraw at any time without consequence, no names will be published, and your answers will be stored securely and used only for this project. You will not be photographed or recorded without separate permission. Name (optional): _____ Signature/date: _____*

Tourist questionnaire (draft). (1) Are you a local or international visitor, and which language do you prefer for information? (2) Which phone do you use, and is there internet in the parks? (3) How often do you photograph wildlife, and do you know the animal's name? (4) How do you identify animals now? (5) What information would you want about an animal? (6) Would you use an identification app, and what would make you trust its result? (7) How would you like to see the result (box, text, AR, voice)? (8) How important is Luganda output?

Guide and ranger interview guide (draft). (1) Which animals do visitors ask about most, and which are most confused? (2) How do you help visitors identify animals now? (3) In what conditions is identification hardest? (4) Would such an app help or compete with your work? (5) Which local names are commonly used? (6) What photography and data rules should the project respect?

Usability evaluation (draft). Tasks: photograph or select an animal photo and read the result; state the species and how sure the system is; read and explain the Luganda text; open a fact sheet; switch language. Questionnaire: the ten-item SUS [36], 5-point Likert items on usefulness, ease of use, trust and enjoyment [30], and open comments. Native Luganda speakers rate each sample text for adequacy and fluency from 1 to 5.

Appendix D Technical/Design Information

Table D.1 lists the seven target classes. Antelope, bird and hog are group classes, and the members shown are examples that will be fixed in the annotation guideline after the existing dataset has been audited.

Table D.1: Target classes.

No.	Class	Scientific name / members
1	Chimpanzee	<i>Pan troglodytes</i>
2	Gorilla	<i>Gorilla beringei beringei</i> (mountain gorilla)
3	Antelope (group)	Bovidae, e.g. Uganda kob (<i>Kobus thomasi</i>), waterbuck, topi, Jackson's hartebeest, bushbuck, eland
4	Hippopotamus	<i>Hippopotamus amphibius</i>
5	Elephant	<i>Loxodonta africana</i>
6	Bird (group)	Class Aves, e.g. species commonly seen by tourists in Uganda (fixed in the guideline)
7	Hog (group)	Suidae, e.g. common warthog (<i>Phacochoerus africanus</i>) and other wild hogs

Luganda content plan. For every class the knowledge base will hold the Luganda name, a short Luganda fact sheet and Luganda sentence templates for four situations: a confident identification, an uncertain identification with alternatives, no animal found, and display of the confidence level. The wording will be drafted in English, translated with model help, and then reviewed, corrected and approved by a native-speaker language specialist, including regional naming variants and grammatical agreement, before use in the application.

Annotation guideline (summary). One tight box per visible animal; label animals that are identifiable and at least about 20 pixels in the smaller dimension; mark tiny, blurred or ambiguous animals “ignore”; box only the visible part of partly hidden animals if at least about 30% is visible; label herd members separately; record distance class, light, occlusion and source for each image; and release only park-level location, never GPS coordinates.

Planned application screens. Home (take photo, upload, language switch); camera/AR view; result view with boxes, class names, scores and Luganda text; class fact sheet; history; and an about/safety page on limitations and responsible wildlife viewing.